

Differences in imaging biomarkers between patients with intermediate and advanced non-neovascular age-related macular degeneration (AMD) in the University of Colorado AMD registry.

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To quantify and compare the different prevalence rates of specific retinal imaging biomarkers in patients with intermediate AMD (iAMD) and advanced non-neovascular AMD (nnAMD). Cross-sectional study of patients with iAMD and advanced nnAMD. Imaging studies were reviewed for qualitative imaging biomarkers. Choroidal thickness measurements were obtained subfoveally and in 1000 μ m and 2000 μ m intervals away from the fovea. The Chi-squared test and Fisher's exact test were used to compare rates of imaging biomarkers among the two cohorts. P-value of <0.05 was considered significant. 376 eyes of 197 patients with iAMD and 187 eyes of 97 patients with advanced nnAMD were recruited. There were significantly lower rates of the following imaging biomarkers in the iAMD compared with the advanced nnAMD cohorts: soft drusen (66.0% vs. 84.2%, $p = 0.001$), calcified drusen (4.3% vs. 40.0%, $p < 0.0001$), RPD (26.2% vs. 53.3%, $p < 0.0001$), ORT (0.5% vs. 46.9%, $p < 0.0001$), RP (1.1% vs. 46.3%, $p < 0.0001$), pigment migration (53.2% vs. 100%, $p < 0.0001$), and iRORA (17.9% vs. 80.2%, $p < 0.0001$). In the iAMD cohort, choroidal thickness was significantly greater at 188 μ m (SD: 60) and 194 μ m (SD: 69), compared to the advanced nnAMD with measurements of 153 μ m (SD: 68), and 161 μ m (SD: 76). This difference was statistically significant ($p < 0.0001$ and $p = 0.0002$). Our results highlight significant differences in imaging biomarkers between both cohorts. Key biomarkers, such as iRORA, RPD, pigment migration, and thinner choroidal thickness, were associated with advanced nnAMD. Identifying these biomarkers early may help target patients who could benefit from new treatments, potentially delaying vision loss.

Blood integrin- and cytokine-producing T cells are associated with stage and genetic risk score in age-related macular degeneration.

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Age-related macular degeneration (AMD) remains a leading cause of vision loss in the geriatric population. There are age-related changes in peripheral blood leukocyte composition, but their significance for AMD remains unclear. We aimed to determine changes in immune cell populations in the blood of AMD patients. A standardized 31-parameter flow cytometry analysis was conducted on peripheral blood mononuclear cells from 59 patients with early and advanced AMD and 39 controls without AMD, all older than 65 years. Fundus photography and optical coherence tomography were used to classify disease stages and a custom genotype array was used to compute an AMD genetic risk score based on 52 AMD disease risk variants (GRS-52). A generalized linear regression model corrected for age, sex, and smoking status revealed that AMD patients showed decreased frequencies of CD4⁺ T helper cell population expressing Integrin Alpha E (CD103) ($P_{adj} = 0.019$). We further noted that early AMD was characterized by increased interleukin-4 (IL-4)-producing CD4⁺ T helper cells ($P_{adj} = 0.013$; <0.001), as well as IL-4-producing cytotoxic CD8⁺ T cells ($P_{adj} = 0.016$; <0.001). Reclassification of samples based on the GRS-52 revealed that IL-17-producing T cells decreased incrementally across GRS-52 categories. In AMD, alterations in peripheral blood leukocyte populations are associated with genetic risk score and disease stage and include specifically IL-4 and IL-17A cytokine-producing and CD103 integrin-expressing T cell populations.

Dry and neovascular "wet" age-related macular degeneration: Upcoming therapies.

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The age-related macular degeneration (AMD) field is witnessing promising advancements in therapeutic options. Breakthrough drugs such as pegcetacoplan and avacincaptad have been FDA-approved for dry AMD, marking a significant development as there were no treatment options until August 2023. While several anti-vascular endothelial growth factor (VEGF) inhibitors have been approved for wet AMD, challenges persist with the need for frequent dosing. New treatments such as gene therapy, cell therapy, WNT pathway agonists, complement inhibitors, and anti-VEGF combination drugs are under development to address these issues. These developments are exciting and hold promise for transforming the field of medicine, offering hope for improved outcomes and enhanced patient care in managing AMD.

Risk factors for age-related macular degeneration: an update.

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Age-related macular degeneration (AMD) is the leading cause of irreversible vision loss in the United States and other western nations. Limited treatment is available, and there are no established means of prevention. The detection of modifiable risk factors is important to suggest preventive behaviors that can reduce disease occurrence or prevent the progression to the late stages of AMD. Results of recent studies suggest that the etiology and pathogenesis of AMD are a complex interaction of genetic and external factors. Although a number of factors seem promising, only age and cigarette smoking are confirmed as increasing AMD risk. Other factors that most likely play a significant role in AMD are nutritional factors, e.g., antioxidants, and hypertension or other underlying atherosclerotic disease processes. The results of the Age-Related Eye Disease Study suggest a moderate beneficial effect of antioxidant, vitamin, and zinc supplementation in reducing progression to severe AMD.

Age-related macular degeneration and the aging eye.

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Age-related macular degeneration (AMD) is an ocular disease that causes damage to the retinal macula, mostly in the elderly. Normal aging processes can lead to structural and blood flow changes that can predispose patients to AMD, although advanced age does not inevitably cause AMD. In this review, we describe changes that occur in the macular structure, such as the retinal pigment epithelium and Bruch's membrane, with advancing age and in AMD. The role of genetics in AMD and age-related changes in ocular blood flow that may play a role in the pathogenesis of AMD are also discussed. Understanding the pathophysiology of AMD development can help guide future research to further comprehend this disease and to develop better treatments to prevent its irreversible central vision loss in the elderly.

[Risk factors for age-related macular degeneration].

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To present the results of examinations of the risk factors for age-related macular degeneration (AMD) carried out in the last 3 years. Our clinical material comprised 119 patients, 69 women and 50 men, aged 43-85, mean age 70 years. Using classification according to the worse eye, there were 20 patients with drusen, 27 with atrophic changes and 72 with exudative form and AMD. The following risk factors were evaluated: age, sex, body mass index, history of general medical conditions, cigarette smoking, sun exposure, family history of AMD and ocular conditions such as iris color, lens opacities, hyperopia, gerontoxon and changes in retinal vessels. The significant relationships were found between the development of AMD and the age of patients, as well as between the advanced forms of AMD and the history of cardiovascular diseases and sclerotic changes in retinal vessels, 87% of examined patients have light iris and 52% body mass index above 26. The studies confirmed the role of age in the development of AMD and indicated cardiovascular disturbances, increased body mass index and light iris as the possible risk factors for AMD that are most worth further studying. The special attention should be also paid to drusen as the risk factor of AMD.

Age-related macular degeneration (AMD): Alzheimer's disease in the eye?

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Age-related macular degeneration (AMD) is a late-onset, neurodegenerative retinal disease that shares several clinical and pathological features with Alzheimer's disease (AD), including stress stimuli such as oxidative stress and inflammation. In both diseases, the detrimental intra- and extracellular deposits have many similarities. Aging, hypercholesterolaemia, hypertension, obesity, arteriosclerosis, and smoking are risk factors to develop AMD and AD. Cellular aging processes have similar organelle and signaling association in the retina and brain tissues. However, it seems that these diseases have a different genetic background. In this review, differences and similarities of AMD and AD are thoroughly discussed.

Retinal Aging Transcriptome and Cellular Landscape in Association With the Progression of Age-Related Macular Degeneration.

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Age is the main risk factor for age-related macular degeneration (AMD), a leading cause of blindness in the elderly, with limited therapeutic options. Here, we analyze the transcriptomic characteristics and cellular landscape of the aging retinas from controls and patients with AMD. We identify the aging genes in the neural retina, which are associated with innate immune response and inflammation. Deconvolution analysis reveals that the estimated proportions of M2 macrophages are significantly increased with both age and AMD severity. Moreover, we find that proportions of Müller glia are significantly increased only with age but not with AMD severity. Several genes associated with both age and AMD severity, particularly C1s and MR1, are strong positively correlated with the proportions of Müller glia. Our studies expand the genetic and cellular landscape of AMD and provide avenues for further studies on the relationship between age and AMD.

ROCK-isoform-specific polarization of macrophages associated with age-related macular degeneration.

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Age is a major risk factor in age-related macular degeneration (AMD), but the underlying cause is unknown. We find increased Rho-associated kinase (ROCK) signaling and M2 characteristics in eyes of aged mice, revealing immune changes in aging. ROCK isoforms determine macrophage polarization into M1 and M2 subtypes. M2-like macrophages accumulated in AMD, but not in normal eyes, suggesting that these macrophages may be linked to macular degeneration. M2 macrophages injected into the mouse eye exacerbated choroidal neovascular lesions, while M1 macrophages ameliorated them, supporting a causal role for macrophage subtypes in AMD. Selective ROCK2 inhibition with a small molecule decreased M2-like macrophages and choroidal neovascularization. ROCK2 inhibition upregulated M1 markers without affecting macrophage recruitment, underlining the plasticity of these macrophages. These results reveal age-induced innate immune imbalance as underlying AMD pathogenesis. Targeting macrophage plasticity opens up new possibilities for more effective AMD treatment.

Oxidative Stress and Antioxidants in Age-Related Macular Degeneration.

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Oxidative stress plays a crucial role in aging-related eye diseases, including age-related macular degeneration (AMD), cataracts, and glaucoma. With age, antioxidant reparative capacity decreases, and excess levels of reactive oxygen species produce oxidative damage in many ocular cell types underling age-related pathologies. In AMD, loss of central vision in the elderly is caused primarily by retinal pigment epithelium (RPE) dysfunction and degeneration and/or choroidal neovascularization that trigger malfunction and loss of photo-sensing photoreceptor cells. Along with various genetic and environmental factors that contribute to AMD, aging and age-related oxidative damage have critical involvement in AMD pathogenesis. To this end, dietary intake of antioxidants is a proven way to scavenge free radicals and to prevent or slow AMD progression. This review focuses on AMD and highlights the pathogenic role of oxidative stress in AMD from both clinical and experimental studies. The beneficial roles of antioxidants and dietary micronutrients in AMD are also summarized.
